Fran Cortés

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Experience

AI Research Fellow IBM Research *Dublin, Ireland* **10/2026 – Current**

- Research on LLM security, studying how vulnerabilities transfer from teacher to student models through knowledge distillation. Funded by the EU MSCA project CLOUDSTARS.

Visiting AI Researcher University of Cambridge (CaMLSys) *Cambridge, UK* **09/2025**

- Academic research in agentic red teaming, targeting LLMs, focused on identifying vulnerabilities and evaluating model robustness, collaborating with Nic Lane's group (CaMLSys Lab, University of Cambridge); collaboration ongoing. Contributed to the Flower federated learning framework: github.com/flwrlabs/flower/issues/6039

Visiting AI Research Engineer IBM Research *Dublin, Ireland* **05/2025 – 08/2025**

- Contributed to academic research and the development of a red teaming system targeting LLMs, focusing on identifying vulnerabilities and evaluating model robustness; collaboration ongoing. github.com/IBM/ares/commit/c0e0b69

AI | Cybersecurity Researcher Universidad de Murcia *Murcia, Spain* **10/2024 – Current**

- AI Researcher Engineer focused on Federated Learning and Cybersecurity applied to IoT, leveraging Large Language Models and advanced AI techniques for vulnerability identification and penetration testing.
- Contributed to academic research and the development of secure, privacy-preserving machine learning systems in collaboration with interdisciplinary teams, under the supervision of Prof. Antonio Skarmeta and Prof. José Luis Hernández Ramos.

Specialist Software Engineer Vodafone *Málaga, Spain* **09/2023 – 10/2024**

- Developed microservices using Java Spring Boot, MongoDB, Apache Kafka and Docker, adhering to agile methodologies and industry best practices to deploy new features into production.
- Led the technical hiring process for new graduate engineers, including designing technical scenarios and interview questions.
- Mentored our intern engineers on their final project: simplifying the DB budget management process across portfolios with Power Apps and building a chatbot as an extension.
- Proposed and designed in Figma a frontend application to track our service orders and make our demos clearer.

Deep Learning Engineer Lead Bemyvega *Murcia, Spain* **12/2022 – 05/2023**

- Led the AI team in developing real-time transcription systems using NLP and deep learning techniques, including LLMs. Analyzed and selected appropriate models for each use case, resulting in high-quality transcriptions.

AI Software Engineer | CEO Co-Founder MiSintaxis *Murcia, Spain* **09/2019 – 09/2023**

- Led the creation, design, development and strategy of the number one app in the world of syntax teaching, with more than 85,000 downloads all around the world.
- Applied Natural Language Processing techniques for Spanish language processing. Fine-tuned BLOOM and GPT-2 models on the AnCora corpus, achieving strong F1 scores for Spanish constituency parsing, and deployed them using Hugging Face, PyTorch, Python and AWS. Designed and implemented an architecture based on AWS services.
- Designed and implemented scalable APIs and background workers to develop this mobile app.
- Used Android Studio with Java and XML at first and then led the migration to Flutter in order to create a multiplatform app.
- Recruited 1 designer and 3 software engineers.
- Used the lean methodology to create a minimum viable product and implemented the agile methodology with Scrum using Jira.
- Managed the business side, including legal issues, branding, recruiting, fundraising, external communications and networking.

Education

Master's Degree in Artificial Intelligence Universidad de Murcia *Murcia, Spain* **09/2025 – 07/2026**

- Major in Artificial Intelligence. Overall GPA of 9.14/10. Master's thesis — an agentic framework for the automated red-teaming of LLMs — graded 9.9/10 (Sobresaliente).

Bachelor of Computer Engineering with Honors Universidad de Murcia *Murcia, Spain* **09/2019 – 07/2023**

- Major in Computer Engineering. GPA of 8.6/10, in the top 1% of the university.

Erasmus — Computer Engineering Politechnika Wrocławska *Wrocław, Poland* **09/2021 – 02/2022**

- Major in Computer Engineering. Studied abroad in Poland for one semester. All classes were in English.

Research

- **Cortés-Delgado, F. J.**, Rodríguez, M., et al. (2026). *ContextualJailbreak: Evolutionary Red-Teaming via Simulated Conversational Priming.* arXiv:2605.02647. Black-box evolutionary search over simulated multi-turn primed dialogue, guided by a graded 0–5 harm score from a two-level judge. Across 50 HarmBench behaviors it reaches 100% ASR on gpt-oss:20B, qwen3-8B, and llama3.1:70B, and 90% on gpt-oss:120B; the 40 most harmful attacks transfer without adaptation to closed frontier models (90% gpt-4o-mini, 70% gpt-5, 70% gemini-3-flash), revealing a pronounced provider-level asymmetry in alignment robustness.
- **Cortés-Delgado, F. J.**, Mármol Campos, E., Hernández-Ramos, J. L., Skarmeta, A., Latif, S., Djenouri, D., Krenn, S., Puiu, A., & Vizitiu, A. (2026). *Enhancing security and privacy in federated learning for distributed systems: The REMINDER approach.* In Proceedings of the Global IoT and Edge Computing Summit (GIECS 2025) (pp. 57–70). Springer. doi.org/10.1007/978-3-032-09555-8_4

Skills

- Natural Language Processing | Deep Learning | Hugging Face | PyTorch | Python
- Java Spring Boot | Docker | JavaScript | HTML | CSS | C++ | C | MongoDB | MySQL | AWS | Git | Scrum | Unit Testing | Cucumber | BDD
- Management | Microservices | Artificial Intelligence | Full-Stack
- English, Spanish, French — All professional proficiency or above

Awards

- **Best Innovative Technology Project Award — Master's Thesis (2026):** Awarded by the Social Council of the Universidad de Murcia (XI edition of the Award for the Best Business Creation Project based on a Final Degree or Master's Thesis) for *AgentShield*, a startup project built on my master's thesis on agentic automated red-teaming of LLMs.
- **Extraordinary Final Degree Project Award:** Received recognition for my final project. Applied Natural Language Processing techniques for Spanish language processing. Conducted fine-tuning of different LLMs. Designed and implemented an architecture based on AWS services.
- **Extraordinary High School Award:** Recognized for passing every subject of my final year of high school with Honors.
- **First place** in multiple startup competitions and regional contests with my startup, MiSintaxis.
- **Awarded scholarships** to take courses in business administration and management.

Others

- **Reviewer:** Workshop on Trustworthy AI for Good (AI4GOOD), co-located with NeurIPS 2026.
- Volunteered as a mentor for international students in my city.
- Gave tech talks at several universities and high schools.